

Time Series Analysis

Seminar 0: Introduction: components and exponential smoothing



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **INS**: Institutul Național de Statistică — the National Institute of Statistics (Romania)
- **MAE**: Mean Absolute Error
- **RMSE**: Root Mean Squared Error
- **STL**: Seasonal and Trend decomposition using Loess
- **TSA**: Time Series Analysis

Today's Question and Seminar Structure

- The questions of the seminar
  - how do we open the course notebooks and load a series?
  - which growth rate tells us how the economy is doing?
  - does today's value tell us anything about tomorrow's?
  - how good is the simplest possible forecast?
- Results to obtain (nothing is handed in)
  - the answers to the A exercises
  - the numbers and the chart asked in each B exercise
  - the corrected code and numbers of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Exercise	Question	Type	Model
A1	growth rates of a quarterly series	Solved	–
A2	a sample autocorrelation by hand	Solved	–
A3	naive forecasts and their errors	Solved	–
B1	Romanian GDP: which growth rate?	Solved	–
B2	EUR/RON: memory of levels and of changes	Solved	–
A4	the same calculations on new numbers	Proposed	A1–A3
B3	Romanian inflation: monthly and annual	Proposed	B1, B2
B4	electricity: naive or seasonal naive?	Proposed	A3
C1	a project idea	Open	–
C2	find the errors in an AI answer	Proposed	A1, A3, B1
After the lecture	C1, step 3: decompose the series with STL (Chapter 0 lecture) and compare the seasonal amplitude over time	Open	–

- **[Solved]**: full solution in the slides and the notebook, a model to follow; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: calculations on paper; Part B: real data in the notebook, each task ending with an interpretation question; Part C: open questions and the critique of an AI answer

Prerequisites for Today (1/11): series, frequency and components

- **Time series** $y_1, \ldots, y_T$: one variable observed at successive moments
  - t: the time index (quarter, month, day); T: the number of observations
  - frequency: m periods a year; quarterly $m = 4$, monthly $m = 12$
- **Components**: the series as a sum of three parts

$$y_t = T_t + S_t + E_t$$

  - T_t: the trend, the slow long-run movement
  - S_t: the seasonal component, a pattern that repeats every m periods, $S_t = S_{t-m}$
  - E_t: the irregular part, what remains
  - unadjusted series contain S_t; seasonally adjusted series show $y_t - S_t$
- **Sample mean** and **standard deviation**

$$\bar{y} = \frac{1}{T}\sum_{t=1}^{T} y_t, \qquad s = \sqrt{\frac{1}{T-1}\sum_{t=1}^{T}(y_t - \bar{y})^2}$$

  - s: the typical distance of a value from $\bar{y}$, in the units of the series (percentage points for a rate in %)

Prerequisites for Today (2/11): growth rates

- **Growth over one period**: quarter on quarter (q/q) or month on month

$$g_t = 100 \left(\frac{y_t}{y_{t-\mathbf{1}}} - 1 \right)$$

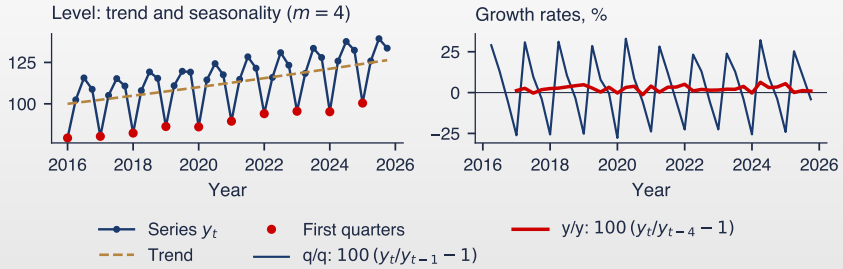
  - $y_{t-\mathbf{1}}$: the value of the previous period; g_t: the change in percent of that value
  - example: $y_{t-\mathbf{1}} = 200$, $y_t = 210$ give $g_t = 100\,(210/200 - 1) = 5\%$
  - for unadjusted data, g_t mixes the season with the economy (spring after winter)

- **Growth over a year**: year on year (y/y)

$$g_t^{(m)} = 100 \left(\frac{y_t}{y_{t-m}} - 1 \right)$$

  - y_{t-m}: the value of the same season one year earlier
  - compares the same season, so the seasonal pattern cancels out
  - for a price index: g_t is the monthly inflation, $g_t^{(\mathbf{12})}$ the annual inflation

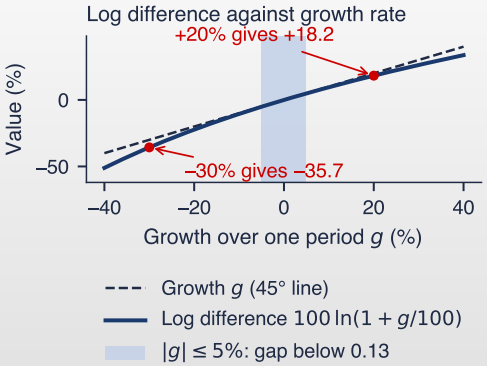
Prerequisites for Today (3/11): a seasonal series and its two growth rates



- Left: a simulated quarterly series, a rising trend (dashed) times a seasonal pattern; the first quarters (red) are always the lowest
- Right: the two growth rates of the same series
  - q/q (blue) swings every year between -28% and $+33\%$: it shows the season
  - y/y (red) stays between -1.4% and $+6.3\%$, around its mean of 2.4%: it shows the underlying growth

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Prerequisites for Today (4/11): log differences and log returns



- **Log difference**: the growth measured on the log scale

$$d_t = 100\,(\ln y_t - \ln y_{t-1}) = 100 \ln\left(1 + \frac{g_t}{100}\right)$$

  - ln: the natural logarithm; g_t: the growth rate of the previous slides
  - $d_t \approx g_t$ for small changes: the gap is below 0.13 for $|g_t| \leq 5\%$
  - the gap grows with the size of the change: -30% gives -35.7, $+20\%$ gives $+18.2$

- **Log return** of a price P_t

$$r_t = 100\,(\ln P_t - \ln P_{t-1})$$

  - log returns add up: the sum of the daily r_t of a month is the log return of the month

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Prerequisites for Today (5/11): the sample autocorrelation

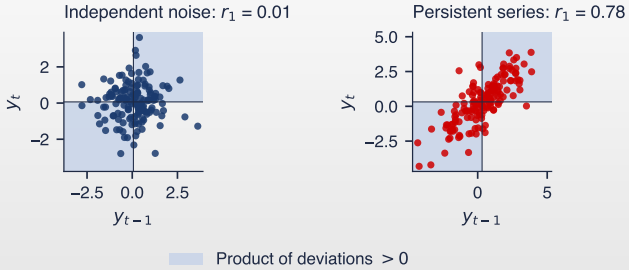
- **Sample autocorrelation** at lag k: how strongly y_t is related to its own value k periods earlier

$$r_k = \frac{\sum_{t=k+1}^{T}(y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{T}(y_t - \bar{y})^2}$$

  - $\bar{y}$: the sample mean; k: the lag, the number of periods between the two values compared
  - numerator: the products of deviations from the mean, positive when y_t and y_{t-k} lie on the same side of the mean
  - denominator: the sum of squared deviations; it keeps r_k between -1 and 1

- **Reading** r_k
  - $r_k > 0$: a high value tends to be followed, k periods later, by a high value
  - $r_k < 0$: a high value tends to be followed by a low value; $r_k \approx 0$: no linear relation
  - r_k^2: about the share of the variance of y_t that a linear forecast from y_{t-k} explains

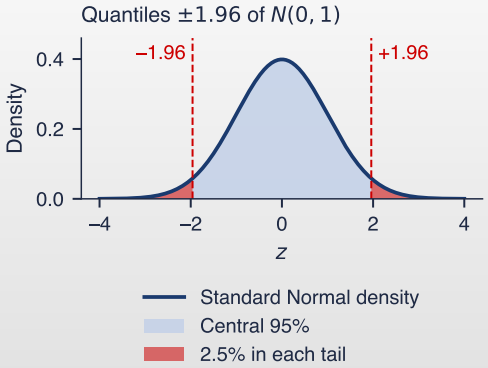
Prerequisites for Today (6/11): the lag plot



- Each point is a pair $(y_{t-\mathbf{1}}, y_t)$; the lines at the mean $\bar{y}$ split the plane into four quadrants
- Shaded quadrants: both deviations have the same sign, their product is positive and raises $r_{\mathbf{1}}$
- Two simulated series, $T = 150$
  - left: independent noise, points in all four quadrants, $r_{\mathbf{1}} = 0.01$
  - right: each value is 0.8 times the previous one plus a new shock; the points lie mostly in the shaded quadrants, $r_{\mathbf{1}} = 0.78$

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Prerequisites for Today (7/11): the ACF and its 95% band



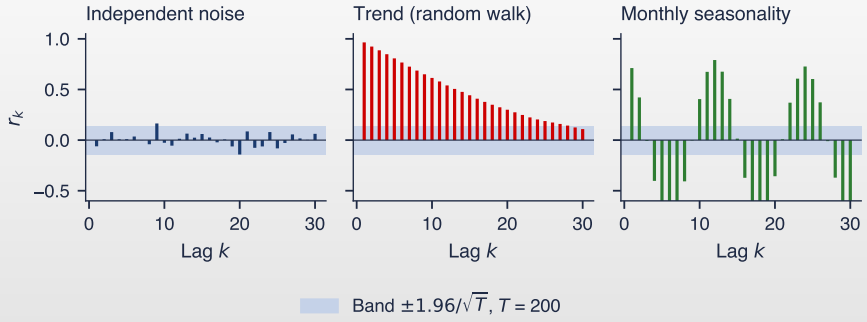
- **ACF** (autocorrelation function): $r_1, r_2, \ldots$ plotted against k
- If the y_t are **i.i.d.** (independent and identically distributed), each r_k is approximately $N(0, 1/T)$; the 95% band is

$$\pm \frac{1.96}{\sqrt{T}}$$

  - $N(0, 1/T)$: the Normal distribution with mean 0 and variance $1/T$
  - 1.96: the 97.5% quantile of the standard Normal distribution; 95% of its probability lies between -1.96 and 1.96
  - $T = 100$: band ± 0.196; $T = 400$: ± 0.098; more data, a narrower band
- A bar outside the band: r_k is significantly different from 0 at the 5% level
  - with 20 lags, about one bar crosses the band by chance (5% of 20)

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Prerequisites for Today (8/11): three typical ACF patterns



- Three simulated series, $T = 200$, lags 1 to 30; shaded: the band $\pm 1.96/\sqrt{T} = \pm 0.139$
- What each panel shows
  - independent noise: only 2 of the 30 bars lie just outside the band, by chance
  - a trend (random walk): $r_1 = 0.96$ and a slow decay ($r_{30} = 0.11$): levels with a trend remember a lot
  - monthly seasonality: $r_{12} = 0.79$ and $r_6 = -0.79$: peaks at the multiples of $m = 12$

Prerequisites for Today (9/11): naive forecasts and the split by time

- **Naive forecast**: the last observed value

$$\hat{y}_{T+h} = y_T, \qquad h = 1, 2, \ldots$$

  - $\hat{y}_{T+h}$: the forecast of the value h periods after the last observation T; h: the horizon
- **Seasonal naive forecast**: the value of the same season one year earlier

$$\hat{y}_{T+h} = y_{T+h-m}, \qquad h \leq m$$

  - for $h > m$, the forecasts repeat the last observed year again
- **Training set and test set**: a split by time
  - training set $y_1, \ldots, y_T$: the forecasts are built only from these values
  - test set $y_{T+1}, \ldots, y_{T+H}$: the next H values, used only to measure the errors
  - a random split would forecast the past with data from the future

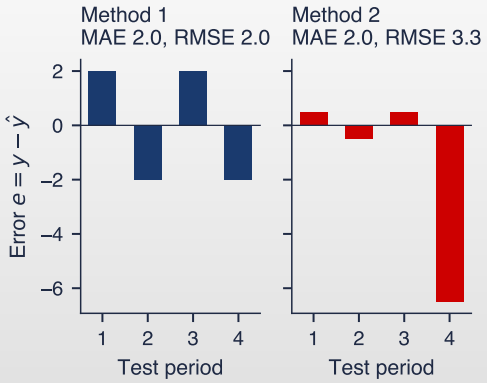
Prerequisites for Today (10/11): naive forecasts of a seasonal series



- A simulated quarterly series: 20 quarters of training (blue), then 4 test quarters (shaded)
- The two forecasts made at the end of the training set
  - naive (red): a flat line at the last value; it misses the low first quarter
  - seasonal naive (green): it copies each quarter of the last training year and follows the test values closely

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Prerequisites for Today (11/11): forecast errors: MAE and RMSE



- **Errors** on the test set: $e_j = y_{T+j} - \hat{y}_{T+j}$, $j = 1, \ldots, H$

$$\text{MAE} = \frac{1}{H}\sum_{j=1}^{H} |e_j|, \qquad \text{RMSE} = \sqrt{\frac{1}{H}\sum_{j=1}^{H} e_j^2}$$

  - MAE (mean absolute error): the average size of the errors
  - RMSE (root mean squared error): squares the errors, so a large error weighs more
  - both are in the units of the series; the smaller, the better
  - always RMSE $\geq$ MAE, with equality only when all $|e_j|$ are equal

- **Chart**: two methods with the same MAE, 2.0
  - method 2 has one large error, so its RMSE is 3.3, against 2.0 for method 1

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Data used today

Series	Source	Unit	Frequency	Exercise
Romanian real GDP	Eurostat, unadjusted	billion EUR, 2010 prices	quarterly	B1, C2
EUR/RON	BNR reference rate	lei per euro	daily, from 2015	B2
Romanian HICP	Eurostat	index, 2015 = 100	monthly, from 2015	B3
Electricity generation	Eurostat	TWh per month	monthly, from 2008	B4

- HICP: Harmonised Index of Consumer Prices; BNR: the National Bank of Romania; TWh: terawatt-hour
- Public sources, read online without a key; the notebook defines one function per source

Setup (1/2): open the notebook and save a copy

- **Question**: can you run Python on the course data in the browser?
- **Inputs**: Google Colab: Google's free notebook service; Python runs on Google's servers, a Google account is enough
- **Tasks**
  1. Open the seminar notebook in Colab.
  2. Choose *File* → *Save a copy in Drive*, so that your changes are kept.
  3. Run the cells of the section *Setup* in order with Shift + Enter; they import the libraries and define the data functions.
  4. If a cell fails, choose *Runtime* → *Restart session* and run again from the top.
- **Report**: nothing; you are ready when the check on the next slide prints the expected numbers

Setup (2/2): load a series

```
# Romanian real GDP, quarterly, unadjusted
g = read_eurostat('namq_10_gdp',
                  'Q.CLV10_MEUR.NSA.B1GQ.RO') / 1000
print(len(g), g.index[0].date(), g.index[-1].date())
print(g.tail(2).round(1))
g.plot()          # a first chart
```

- `read_eurostat(dataset, key)` reads one Eurostat series; the key lists the dimensions (frequency, unit, adjustment, indicator, country)

- **Expected output**
  - 126 quarters, from 1995 Q1 to 2026 Q2
  - last value: 45.88 billion EUR
- Eurostat revises its data: a slightly different last value is normal
- If the count differs a lot, check the key before going on

A1 [Solved]: Growth rates — task

- **Question**: did the Romanian economy collapse in the first quarter of 2025?
- **Inputs**: real GDP, billion EUR: 38.7, 45.6, 52.8, 59.2 (2024 Q1–Q4), 38.9, 46.8 (2025 Q1–Q2)
- **Tasks**
  1. Compute the quarter-on-quarter growth rates for 2024 Q2 to 2025 Q2.
  2. Compute the year-on-year growth rates for 2025 Q1 and 2025 Q2.
  3. Compute the log difference for 2025 Q1 and compare it with the quarter-on-quarter rate.
- **Report**: the two kinds of growth rates, and one sentence that answers the question
- Notebook: section A1

Back Exercise list

A1 [Solved]: Solution

Quarter	y_t	q/q, %	y/y, %
2024 Q1	38.7	–	–
2024 Q2	45.6	+17.8	–
2024 Q3	52.8	+15.8	–
2024 Q4	59.2	+12.1	–
2025 Q1	38.9	−34.3	+0.5
2025 Q2	46.8	+20.3	+2.6

- q/q: quarter on quarter; y/y: year on year

- **Step by step**
  - 2025 Q1, q/q: $100\,(38.9/59.2 - 1) = -34.3\%$
  - 2025 Q1, y/y: $100\,(38.9/38.7 - 1) = +0.5\%$
  - log difference: $100\,(\ln 38.9 - \ln 59.2) = -42.0\%$, far from -34.3%: the approximation fails for large changes
- **Interpretation**
  - no collapse: the fall of a third from Q4 to Q1 happens every winter; compared with a year earlier, GDP grew by 0.5%

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A2 [Solved]: An autocorrelation by hand — task

- **Question**: is a high value usually followed by a high value in this short series?
- **Inputs**: $y = 2, 4, 3, 5, 4, 6$ ($T = 6$)
- **Tasks**
  1. Compute the mean $\bar{y}$ and the deviations $y_t - \bar{y}$.
  2. Compute the sum of squared deviations and the sum of products of consecutive deviations.
  3. Compute r_1 and r_2, and compare them with the 95% band (Prerequisites 7/11).
- **Report**: $\bar{y}$, r_1, r_2, the band, and one sentence on what can be concluded from 6 observations
- Notebook: section A2

Back Exercise list

A2 [Solved]: Solution

t	y_t	$d_t = y_t - \bar{y}$	$d_t d_{t-1}$
1	2	-2	–
2	4	0	0
3	3	-1	0
4	5	1	-1
5	4	0	0
6	6	2	0
sum	24	0	-1

- $\bar{y} = 24/6 = 4$; $\sum d_t^2 = 4 + 0 + 1 + 1 + 0 + 4 = 10$
- $r_1 = -1/10 = -0.10$
- lag 2: $\sum d_t d_{t-2} = (-1)(-2) + (1)(0) + (0)(-1) + (2)(1) = 4$, so $r_2 = 0.40$
- band: $\pm 1.96/\sqrt{6} = \pm 0.80$
- **Interpretation**
  - both values lie well inside the band: 6 observations are far too few to detect memory

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A3 [Solved]: Naive forecasts and their errors — task

- **Question**: for a seasonal series, which simple forecast is better: the last value or the same quarter last year?
- **Inputs**: quarterly sales; training set: 10, 14, 18, 12 (year 1) and 11, 15, 20, 13 (year 2); test set: 12, 16, 21, 14 (year 3)
- **Tasks**
  1. Write the naive forecasts and the seasonal naive forecasts ($m = 4$) for the four quarters of year 3.
  2. Compute the errors $e = y - \hat{y}$ of each method.
  3. Compute the MAE and the RMSE of each method.
- **Report**: a table with the forecasts and errors, the four error measures, and one sentence naming the better method
- Notebook: section A3

A3 [Solved]: Solution

	Q1	Q2	Q3	Q4
actual (year 3)	12	16	21	14
naive	13	13	13	13
error	-1	3	8	1
seasonal naive	11	15	20	13
error	1	1	1	1

- **Naive**: every forecast is the last value, 13
  - MAE $= (1+3+8+1)/4 = 3.25$; RMSE $= \sqrt{(1+9+64+1)/4} = 4.33$
- **Seasonal naive**: each quarter of year 2
  - MAE $= 1.00$; RMSE $= 1.00$
- **Interpretation**
  - the seasonal naive forecast is better: it keeps the seasonal pattern; the large Q3 error inflates the naive RMSE

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B1 [Solved]: Romanian GDP — Task

- **Question**: which growth rate should be reported for an unadjusted quarterly series?
- **Inputs**: Romanian real GDP, unadjusted, 1995 Q1 – 2026 Q2 (Eurostat)
- **Tasks**
  1. Compute the quarter-on-quarter and the year-on-year growth rates of the whole series.
  2. Plot the two growth rates since 2010 on the same chart.
  3. Count the negative values of each rate, and compute the average q/q growth of the first quarters.
  4. Interpretation: which of the two rates shows the recessions?
- **Report**: the counts of negative values, the average Q1 growth, the latest two rates, and the answer to the interpretation question
- Notebook: section B1

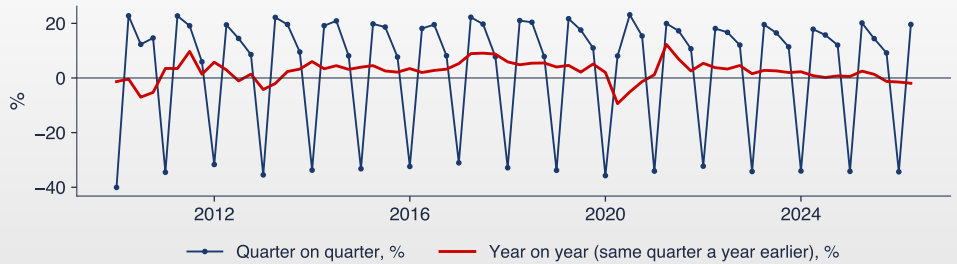
Back Exercise list

B1 [Solved]: Solution (1/2) — step by step

- **Quarter on quarter**: 125 values, 32 negative
  - the first quarters average −36.3%: a fall of a third every winter
  - standard deviation 24.6 percentage points: dominated by the season
- **Year on year**: 122 values, 28 negative
  - negative in the years 1997, 1998, 1999, 2000, 2001, 2009, 2010, 2012, 2013, 2020, 2025, 2026
  - average 2.9%, standard deviation 4.9 percentage points
- **Latest quarter**, 2026 Q2
  - q/q: +19.6% (spring after winter); y/y: −1.9%

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B1 [Solved]: Solution (2/2) — chart and interpretation



- Blue: q/q swings between about −35% and +20% every year; red: y/y moves slowly, around the business cycle
- **Interpretation**: only the y/y rate shows the recessions (2009–2010, 2020) and the slowdown of 2025–2026; for unadjusted data, report y/y growth

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B2 [Solved]: EUR/RON: memory of levels and changes — task

- **Question**: does today's exchange rate help to forecast tomorrow's, and does today's change help to forecast tomorrow's change?
- **Inputs**: EUR/RON reference rate of the BNR, 5 January 2015 – 18 September 2026 (2 939 days)
- **Tasks**
  1. Compute the daily log returns r_t (Prerequisites 4/11).
  2. Compute the sample ACF of the level and of the returns for lags 1–20.
  3. Compare r_1 of the returns with the 95% band (Prerequisites 7/11), and count the lags outside the band.
  4. Interpretation: is the naive forecast a sensible benchmark for the exchange rate?
- **Report**: r_1 and r_{20} of the level, r_1 and r_2 of the returns, the band, the number of lags outside it, and the answer to the interpretation question
- Notebook: section B2

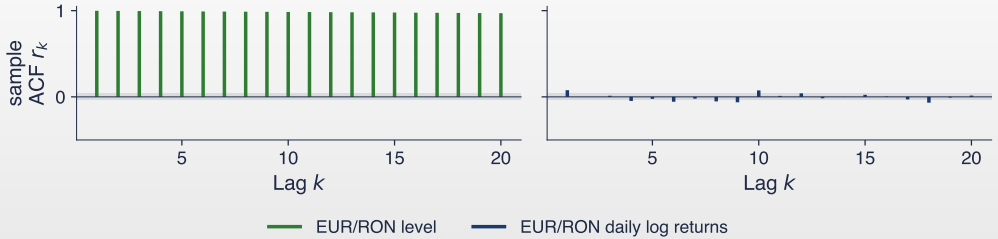
Back Exercise list

B2 [Solved]: Solution (1/2) — step by step

- **Level**: $r_1 = 1.00$, $r_{20} = 0.97$
  - the level remembers itself for weeks: a slow decay, the signature of a trend
- **Returns**: standard deviation 0.13% a day; $r_1 = 0.078$, $r_2 = -0.0003$
  - band ± 0.036; 8 of the 20 lags lie outside it
  - r_1 is significant but small: yesterday's change explains less than 1% of the variance of today's change (r_1^2)

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B2 [Solved]: Solution (2/2) — chart and interpretation



- Left: the ACF of the level stays close to 1; right: the ACF of the returns is almost flat; shaded: the band $\pm 1.96/\sqrt{T}$
- **Interpretation**: yes; the level of today is the best simple forecast of tomorrow's level, while past changes add very little: the naive forecast is the benchmark to beat

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A4 [Proposed]: New numbers — task

- **Question**: can you repeat A1–A3 without the solution in front of you?
- **Inputs**: a quarterly series 120, 150, 132, 168, 126 (Q1 of year 1 to Q1 of year 2); a series 5, 7, 6, 8, 9; a series with $m = 3$: training 20, 30, 25, 22, 32, 27, test 23, 33, 29
- **Tasks**
  1. Model: A1, A2 and A3 [Solved].
  2. Compute the period-on-period growth rates, the log differences and the year-on-year rate of the last quarter.
  3. Compute r_1 of the series 5, 7, 6, 8, 9 and the 95% band (Prerequisites 7/11).
  4. Compute the MAE and the RMSE of the naive and seasonal naive forecasts of the test values.
- **Report**: the growth table, r_1 and the band, the four error measures, and one sentence naming the better forecast
- Notebook: section A4

B3 [Proposed]: Romanian inflation — task

- **Question**: does monthly inflation have a seasonal pattern that the annual rate hides?
- **Inputs**: Romanian HICP, 2015 = 100, monthly, 1 January 2015 – 1 August 2026 (Eurostat)
- **Tasks**
  1. Model: B1 and B2 [Solved].
  2. Compute the monthly (m/m) and the annual (y/y) inflation rates in %.
  3. Compute the average monthly inflation for each calendar month.
  4. Compute the ACF of the monthly rate for lags 1–24, and of the annual rate at lag 1.
  5. Interpretation: why is the annual rate so persistent?
- **Report**: the latest monthly and annual rates, the months with the highest and lowest average, r_1 and r_{12} of the monthly rate, r_1 of the annual rate, and the answer to the interpretation question
- Notebook: section B3

Back Exercise list

B4 [Proposed]: Electricity: naive or seasonal naive? — task

- **Question**: for monthly electricity generation, does the seasonal naive forecast beat the naive one?
- **Inputs**: net electricity generation in Romania, TWh per month (Eurostat); test set: August 2024 – July 2026 (24 months)
- **Tasks**
  1. Model: A3 [Solved].
  2. Split the series by time: the last 24 months form the test set.
  3. Forecast the test set with the naive and with the seasonal naive method ($m = 12$), using only the training set.
  4. Compute the MAE and the RMSE of each method, and plot both forecasts against the test data.
  5. Interpretation: why does the seasonal naive method not win clearly here?
- **Report**: the four error measures, the chart, and the answer to the interpretation question
- Notebook: section B4

C1: A project idea

- **Question**: how much of the variation of a Romanian monthly series is seasonal, and is the seasonal pattern changing?
- **Candidate series** (Eurostat or INS)
  - electricity generation, tourist arrivals, retail trade, industrial production
- **Steps**
  - plot the series and its year-on-year growth
  - compare the average of each calendar month in the first and in the last five years
  - after the lecture: decompose the series with STL and compare the seasonal amplitude over time
- A good first step for the team project of Chapters 0 and 4

C2 [Proposed]: Critique an AI answer — the answer

- **Prompt** sent to an AI assistant: "From the quarterly Romanian real GDP, compute the average annual growth rate and the RMSE of a naive forecast."

```
g = read_eurostat('namq_10_gdp', 'Q.CLV10_MEUR.NSA.B1GQ.RO') / 1000
growth = g.diff()                                  # annual growth rate, %
print('average annual growth:', growth.mean())
test = g.sample(frac=0.2, random_state=1).sort_index()   # test set
train = g.drop(test.index)                                # training set
fc = train.reindex(g.index).ffill().shift(1).reindex(test.index)  # naive
rmse = ((test - fc) ** 2).mean()
print('RMSE of the naive forecast:', rmse)
```

- **Conclusion of the AI**: "Romanian GDP grows by only 0.24% a year, and the naive forecast has an RMSE of 111.9 billion EUR, larger than GDP itself: forecasting GDP is hopeless."

Back Exercise list

C2 [Proposed]: Critique an AI answer — task

- **Question**: can you trust code that runs without an error message?
- **Inputs**: the code and the conclusion on the previous slide; the definitions of the primer
- **Tasks**
  1. Model: A1, A3 and B1 [Solved].
  2. Find the three errors planted in the code.
  3. For each error, say how you would detect it: a check on the output, the primer, or a solved exercise.
  4. Write the corrected code, with the last 8 quarters as the test set, and report the corrected numbers.
  5. Say whether the conclusion "forecasting GDP is hopeless" survives.
- **Report**: one line per error in the format: request, answer, error, how found, correction; the corrected growth and RMSE
- Notebook: section C2

Recap, the Chapter 0 lecture and practice at home

- **Today**
  - for unadjusted data, report growth over a year, not over one quarter
  - r_k: how much a series remembers its value k periods earlier
  - levels with a trend remember a lot; daily changes of prices remember little
  - the naive and seasonal naive forecasts are the benchmarks; split by time, then measure MAE and RMSE
  - AI code can run and still be wrong
- **In the lecture**
  - How is the course organised and graded?
  - Which components make up a time series?
  - How does exponential smoothing forecast?
  - Which forecasts win on real Romanian data?
- **At home**
  - repeat B1 for the euro area GDP (key `Q.CLV10_MEUR.NSA.B1GQ.EA20`) and compare the seasonal swings with Romania
  - take the Chapter 0 quiz on the course website: 20 questions, each answer explained